Zane R. Jobe, Ph.D.

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Summary

Sedimentologist with 20+ years of experience in clastic and mixed-lithology deep-water systems. I integrate **outcrop analogs**, **digital outcrop models**, core, subsurface datasets, and modern-seafloor observations to characterize reservoir architecture and predict fluid flow. Director of the Geology Center of Research Excellence (CoRE) at Mines; 60 peer-reviewed publications; h-index 26.

Education

- **Ph.D., Geology**, Stanford University, 2010
  Thesis: Multi-Scale Architectural Evolution and Flow Property Characterization of Channelized Turbidite Systems
- **B.S., Geology (Summa Cum Laude)**, University of Texas at Arlington, 2004

Appointments

- **Associate Research Professor**, Colorado School of Mines, 2016–present
  Director, Geology Center of Research Excellence (CoRE), 2016–present
  Director, Earth Resource Data Science graduate certificate, 2021–present
  Director, PTTC Rockies, 2025–present
- **Research Sedimentologist**, Shell Oil Company, 2010–2016

Research & Funding (selected)

- **$6.5M+** total fundraising at Mines since 2016 (major grants + student/small grants)
- Lead investigator on industry-funded outcrop-analog programs in the Permian Basin, Guadalupe Mountains, Sacramento Mountains, and Qazaqstan carbonate slopes
- **Brushy Canyon Redux (2026–2030):** digitizing 495 legacy measured sections and photopanels; new digitally-native outcrop acquisition in Guadalupe Mountains National Park; outcrop-to-subsurface correlation workflows (outcropanalog.com)
- Published 60 peer-reviewed articles and >100 conference abstracts; h-index 26, i10-index 38

Selected Publications (*student, **postdoc)

Youngest first. Italic notes highlight outcrop and virtual-outcrop contributions.

2026

- Saifudin, L., Jobe, Z.R., Slootman, A., Carr, M., Plink-Bjorklund, P., 2026, Supercritical-flow structures in a Cretaceous submarine channel-lobe transition zone, Point Loma Formation, California. *The Depositional Record*, v. 2(3), e70070. PDF
  Outcrop: coastal-cliff exposures at Cabrillo National Monument; bed-scale architecture mapped in the field for subsurface channel-lobe transition prediction.
- Nworie, C.D., Jobe, Z.R., Slootman, A., Venturelli, R., Eggenhuisen, J.T., Reijmer, J.J.G., 2026, Carbonate turbidity currents play an underappreciated role in the global carbon cycle. *The Sedimentary Record*, 24(1). PDF
  Links experimental and natural carbonate grain datasets to deep-marine sediment routing; informs interpretation of carbonate slope outcrops and cores.

2025

- Simabrata, H., Jobe, Z.R., Hou, P., Wood, L.J., *Kus, K.B., Hurd, G., 2025, Grain-fabric alignments in a mixed carbonate–siliciclastic mass-transport complex: The Cutoff Formation, Permian Basin, Texas. *Sedimentology*, 72 (4), 1065–1101. PDF
  Outcrop: Guadalupe Mountains (McKittrick Canyon, Walnut Canyon); field mapping and sampling of MTD microfabrics as analog for Permian Basin subsurface.
- Simabrata, H., Jobe, Z.R., Wood, L.J., *Cardona, S., Hou, P., 2025, Characterization, prediction and implications of compartments in a mixed carbonate–siliciclastic mass-transport complex: The Cutoff Formation, Permian Basin, Texas. *Sedimentology*, 72 (4), 1102–1131. PDF
  Outcrop: multi-km transects in Guadalupe Mountains; compartmentalization documented at bed-to-landslide scale for reservoir-model parameterization.
- Putri, S.H., Jobe, Z.R., Melick, J., Wood, L.J., French, M., 2025, Depositional-process controls on chemofacies in mixed-lithology submarine lobe deposits: Permian Wolfcamp XY Formation, Delaware Basin, Texas. *Journal of Sedimentary*

Research, 95 (1), 63–85. PDF
Core-focused study paired with regional outcrop context from Delaware Basin analogs (Bone Spring, Cutoff) for lateral facies prediction.

- Musso, M.S., Slootman, A., Wood, L.J., Jobe, Z.R., 2025, Stalked crinoids: from skeleton to sediment. *Journal of Sedimentary Research*. PDF
Outcrop: Sacramento Mountains carbonate fan deposits; grain-size and fabric measured in field and lab to constrain transport dynamics.

2024

- Jobe, Z.R., *Seckinger, S., *Martin, T., *Kus, K., *Pettinga, L., 2024, Lateral heterogeneity of basin-plain turbidites of the Cloridorme Formation, Quebec, Canada. *The Depositional Record*, 11 (1), 22–43. PDF
Outcrop: 10+ km of basin-plain exposures along the St. Lawrence River; measured sections and photopanels used to quantify lateral heterogeneity for horizontal-well prediction.
- Jobe, Z.R., Hubbard, S.M., Romans, B.W., 2024, How many turbidity currents pass through a submarine channel during its lifespan? *Journal of Sedimentary Research*, 94 (6), 737–749. PDF
Synthesizes outcrop (Magallanes Basin, Chile), subsurface, and experimental datasets to estimate turbidity-current frequency through channel lifetimes.

2023

- Gilbert, J.C., Jobe, Z.R., 2023, Submarine-channel element architecture in strike and dip views: Miocene Modelo Formation, Lake Piru, California. *The Sedimentary Record*, 21 (1). PDF
Outcrop: lake-shore and trail exposures at Lake Piru; 3D architectural elements mapped for comparison to subsurface channel models.
- Bishop, J., Bachtel, S., Thompson, J., Miller, C., Ryan, B., Sullivan, M., Jobe, Z.R., 2023, Carbonate submarine fan deposits of the Mississippian Lake Valley Formation, Sacramento Mountains, New Mexico. *The Depositional Record*. PDF
Outcrop: Tierra Blanca anticline and surrounding exposures; fan-lobe architecture documented as analog for carbonate deep-water reservoirs.
- Gross, E., Carr, M., Jobe, Z.R., 2023, Three-dimensional bounding surface architecture of a wet aeolian system: Entrada Sandstone, Utah. *Sedimentology*, 70, 145–178. PDF
Outcrop: drone-assisted photogrammetry and field mapping in Arches NP region; virtual outcrop models used to quantify lateral facies heterogeneity.

2022

- Kus, K.B., Jobe, Z.R., Laugier, F., Walker, W., Sullivan, M., 2022, Quantifying lateral heterogeneity of distal submarine lobe deposits, Point Loma Formation, California. *The Depositional Record*, 8, 472–501. PDF
Outcrop: Cabrillo National Monument cliff sections; statistical analysis of bed-scale heterogeneity for subsurface lobe prediction.
- Martin, T., Tadla, J., Jobe, Z.R., 2022, Digitalization of legacy datasets and machine learning regression: Lewis Shale, Wyoming. *The Sedimentary Record*, 20. PDF
Virtual outcrop / digital: digitized legacy core photos and graphic logs; ML regression workflow applicable to outcrop-section databases.
- Hou, P., Jobe, Z.R., Wood, L., 2022, Statistical characterization of a confined submarine fan system: Lower Atoka Formation, Ouachita Mountains, USA. *Sedimentology*, 69, 775–797. PDF
Outcrop: Pennsylvanian fan deposits in Oklahoma–Arkansas; measured sections and photopanels for confined-fan statistics.

2021

- Walker, W., Jobe, Z.R., Sarg, J.F., Wood, L., 2021, Progradational slope architecture in outcrops of the Bone Spring Formation, Permian Basin, west Texas. *Geosphere*, 17, 1268–1293. PDF
Outcrop: Delaware Basin road cuts and Guadalupe Mountains transects; mixed carbonate–siliciclastic slope documented as producing-zone analog.
- Fryer, R.C., Jobe, Z.R., Laugier, F., Pettinga, L.A., Gilbert, J.C., Shumaker, L.E., Smith, J.E., Sullivan, M., 2021, Submarine lobe deposits of the Point Loma Formation, California. *The Depositional Record*, 7, 374–391. PDF
Outcrop: quantitative bed-scale architecture from Cabrillo exposures; foundation dataset for Point Loma virtual-outcrop and ML facies work.
- Jobe, Z.R., Howes, N., Martin, J., Meyer, R., Coutts, D., Hou, P., Stright, L., Laugier, F., 2021, Sedimentary graphic logs: A template for description and a toolkit for digitalization. *The Sedimentary Record*, 19. PDF
Virtual outcrop: open-source Python toolkit (pylogit) for standardizing and digitizing field and core graphic logs; underpins digital outcrop-model attribute extraction.
- Englert, R.G., Hubbard, S.M., Cartigny, M.J., Clare, M.A., Coutts, D.S., Hage, S., Hughes Clarke, J., Jobe, Z.R., Lintern, D.G., Stacey, C., Vendettuoli, D., 2021, Quantifying the 3D stratigraphic expression of cyclic steps by integrating seafloor

and deep-water outcrop observations. *Sedimentology*. PDF
Outcrop + virtual: Nanaimo Group (BC) and Monterey/Squamish exposures integrated with modern seafloor data; 3D photogrammetry where cliff access permits.

2018–2010 (foundational outcrop studies)

- Jobe, Z.R., Howes, N.C., Romans, B.R., Covault, J.A., 2018, Volumes and recurrence of submarine fan-building turbidity currents. *The Depositional Record*, 4, 160–176. PDF
Outcrop: Magallanes Basin (Chile) channel and fan deposits calibrated against modern Niger Delta slope datasets.
- Malkowski, M.A., Jobe, Z.R., Sharman, G.R., Graham, S.A., 2018, Down-slope facies variability within deep-water channel systems: Cerro Toro Formation, southern Patagonia. *Sedimentology*. PDF
Outcrop: Cerro Toro channel-fill exposures (Sierra del Toro, Chile); bed-thickness trends along dip documented over km scales.
- Jobe, Z.R., Bernhardt, A., Lowe, D.R., 2010, Facies and architectural asymmetry in a conglomerate-rich submarine channel fill, Cerro Toro Formation, Magallanes Basin, Chile. *Journal of Sedimentary Research*, 80 (12), 1085–1108. PDF
Outcrop: PhD field area in southern Chile; quantitative architectural analysis of asymmetric channel fills that remains a standard deep-water analog.

Full publication list: zanejobe.github.io/publications

Leadership, Mentoring & Field Programs

- Lead a research group of **9 FTEs**; advise **7** current MSc/PhD students (outcrop-based projects in TX, NM, CA)
- Since 2016: **13 MS** and **4 PhD** graduates; **3** postdocs; **28** undergraduates with research experience
- **40+ industry and academic field trips** led since 2010 (Texas, California, Chile, Ireland, New Mexico, Spain, Brazil)
- **Virtual field trips:** four-module deepwater sedimentology course for BP (2022–2024); digital outcrop-model and ML workflows with graduate and CS capstone teams
- **2025** Invited Keynote, Digital Geoscience conference, Geological Society of London
- **2019–2021** SEPM Sedimentology Councilor (elected)
- **2017** Keynote, Deep-water Depositional Systems (Geological Society of London)

Teaching (selected)

- Founder & Director, Earth Resource Data Science online graduate certificate
- Graduate courses: GEOL 557/558 (Python data analysis); GEOL 624 Carbonate Sedimentology; seminars on turbidite architecture and core interpretation
- **550+ student credit hours** taught since 2016 (online and classroom)

Fieldwork Experience (cumulative)

Region	Weeks	Primary focus
Chile (Magallanes Basin)	29	Deep-water channel and fan outcrops
California	24	Submarine channels, lobes, slope systems
Texas / New Mexico	37	Permian deep-water and carbonate slopes
Ireland (Clare Basin)	12	Turbidite and MTD outcrops
Utah	8	Eolian and mixed-system analogs

Selected Invited Presentations

- May 2026 — Keynote, Indonesia Petroleum Association (IPA), Pertamina pavilion
- Oct 2025 — Keynote, Digital Geoscience, Geological Society of London
- Dec 2024 — Plenary, BSRG Annual Meeting
- Jan 2017 — Keynote, Deep-water Depositional Systems, Geological Society of London

Professional Memberships

SEPM · IAS · AAPG · AGU · GSA